Thiago Mata

Technology Leader

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Technology leader who connects engineering to business outcomes. Over twenty years across startups, ad-tech at billions of transactions, and regulated banking have shaped how I choose what to build, what not to build, and where to invest, and how I lead teams that decide well today and better tomorrow, always favouring clarity over cleverness and reliability over novelty.

My background is in distributed systems, real-time data, and cloud platforms, from public systems used by every university in Brazil to bot detection at the edge of a global ad-tech platform. Alongside that work, I publish research on formal verification, using machine-checked proofs to show software is correct rather than only testing it, which shapes how I think about reliability.

This is the full version of my CV. For a summary, see the one-page CV →

EXPERIENCE

Senior Manager, Macquarie Bank

Sydney · 2024–Present

- Lead a new, high-performing team with strong retention and fast onboarding, setting team and individual goals together and tracking progress through regular evaluations.
- Delivered new banking products while reshaping how the team builds them, cutting production incidents through testing, observability, graceful degradation, and clearer code.
- Built data services that combine real-time events and daily snapshots into a single, consistent view of customer accounts for Banking and Financial Services.
- Contributed to a new digital term deposit experience that reduced account opening from days to minutes.
- Interview engineering candidates as part of the hiring panel.

Java, Kafka, gRPC, Protobuf, BigQuery, Apache Beam, Dataflow, Spanner, Pub/Sub, Cloud Functions, Elasticsearch, Dynatrace, Docker, Helm, Kubernetes Case study →

Senior Software Engineer, Rokt

Sydney · 2019–2024 · 5 years 5 months

- Led the Bot Detection team within Traffic Resilience: a real-time service on the critical path of all company traffic, designed for graceful degradation so it would never go down or block legitimate users. It combined real-time and offline analysis, rate limiting, local residency enforcement, and anomaly detection on Kubernetes edge clusters.
- Built the Identity System, which groups customer interactions for behaviour analysis, and the Attribution System, which links interactions to advertiser conversions and feeds campaign metrics to machine learning models.
- Worked on the Customer Data Platform, a real-time distributed system that cleans, validates, and enriches customer data, and later on anomaly detection for key business metrics.
- Interviewed engineering candidates as part of the hiring panel.

Java, Scala, Python, Go, Spark, Kafka, Gremlin, Neptune, Redshift, DynamoDB, SQS/SNS, EKS, CloudFormation, Terraform, Elasticsearch, Redis, HBase, Grafana, Prometheus, Datadog, Buildkite, Helm, Istio, Envoy, gRPC Bot detection → Identity → Attribution →

Senior Data Engineer, SLI Systems

Christchurch · 2017–2019 · 1 year 8 months

- Migrated core data services from Apache Spark to Apache Beam on Google Cloud, making them faster and cheaper and strengthening the company's position in its acquisition.
- Turned data scientists' models into production SQL and Spark Streaming jobs, and measured the impact and cost of each new feature through online and offline tests.

Java, Python, BigQuery, Apache Beam, Dataflow, Cloud Storage, Spanner, Pub/Sub, Elasticsearch, Jenkins, Docker, Helm, Kubernetes Case study →

Chief Technology Officer, Ideia Big Data

Brasília · 2014–2017 · 3 years

- Led engineering across three offices in three states, taking the company's research, microtargeting, and statistics products from concept to their first production-ready versions.
- Defined the technical strategy and reported it to investors; the products combined public demographic data with distributed systems, machine learning, and cloud streaming.
- Hired the engineering team and created its career ladder, defining the responsibilities and expected competencies for each level, along with the training policy and development process that prepared the company for its merger with Pinion.
- Negotiated with cloud providers and led the migration to a new one, cutting infrastructure costs while improving scale, reliability, and observability.
- Validated features with clients through engagement and feedback.

Python, R, Scala, PHP, Node.js, Spark, Hadoop, Cloudera, Postgres, MongoDB, Redis, Elasticsearch, AWS, Docker, Grafana, Angular, D3.js Case study →

Senior Programmer, Adjetiva

Brasília · 2014 · 10 months

Part of the team behind SAT-JF14, a mobile app that won three Cannes Lions. Also built the Embratur brand channel, mobile apps for World Cup events, educational games, and the Fiat EcoDrive desktop app.

Java, Node.js, Angular, PHP, Laravel, Postgres, Cordova, Phaser, Scrum

Co-Founder, OtimiCar

Brasília · 2013–2014 · 4 months

Specified and built the first server for a device that turns any car into a smart car, syncing car data with partners, sending maintenance recommendations to owners, and reporting to partners on their customers and products.

Java, Node.js, MongoDB, Postgres, REST, SOAP, IoT Case study →

Senior Coordinator, Grupo 954

Brasília · 2012–2013 · 1 year 6 months

Led the engineering team, interviewed candidates, and modelled the company's estimation, specification, and development processes, building web applications for data analysis clients, including General Motors.

PHP, Oracle, SQL Server, REST, UML, BPMN, Scrum

Process Analyst, INEP

Brasília · 2009–2012 · 3 years 1 month

Led a team of engineers on high-availability, high-volume systems with many concurrent users, including E-MEC, which handles accreditation and quality assurance for every university in Brazil, and Revalida, the national exam for revalidating foreign medical diplomas.

PHP, Zend, Laravel, Java, JBoss Seam, Oracle, Redis, Memcached, BPMN, Scrum Case study →

EARLIER EXPERIENCE

Analyst and Web Developer, SECOM, Presidency of Brazil

Brasília · 2008–2009 · 7 months

Requirements, modelling, prototyping, and web development. PHP, Zend, Oracle, Postgres

Database Administrator, Bancoob

Brasília · 2007–2008 · 1 year 2 months

Built the credit portfolio transfer system for a high volume of financial transactions and ran the accounting close, using stored procedures, memory tables, and bulk loading. SQL Server, Visual Basic, Java

Web Developer, Ministry of Education

Brasília · 2006–2007 · 1 year

Built the professional education supervision system and the integrated planning, budget, and finance system, with fast reports over real-time and batch data. PHP, Zend, Postgres, SVG

Senior Developer, AgênciaClick/Isobar

Brasília · 2005–2006 · 1 year

Built SISAC, a system for monitoring the Presidency of Brazil's communication actions, plus the BrazilTour site and the Atlantic Forest NGO Network site. PHP, Postgres, Oracle

Developer, SAMU

Brasília · 2005 · 6 months

Front-end developer on the ambulance request system of an emergency call centre that connects patient, ambulance crew, and medical team in one call. Java, Struts, XUL

Teacher, Studio On-Line — Brasília · 2004–2010 · 6 years

Taught web development. PHP, MySQL, HTML, CSS, JavaScript

Developer, M.S.D. — Brasília · 2003–2004 · 11 months

Built computer-based training. Authorware, Flash, ToolBook, JavaScript

SKILLS

Leadership	Technical strategy, hiring and interviewing, people management, goal setting and performance reviews, investor reporting, vendor negotiation, distributed teams, career paths, mentoring, process design, cost optimisation, reliability of critical real-time systems, risk and compliance in regulated environments
Languages	Scala, Java, Python, Go, JavaScript, Node.js, PHP, R, C#, Haskell
Data	Spark, Apache Beam, Hadoop, Kafka, BigQuery, Redshift, Athena, EMR, Bigtable, Spanner
Databases	Postgres, MySQL, Oracle, SQL Server, MongoDB, DynamoDB, Neptune, HBase, Redis, Elasticsearch
Cloud	AWS (EC2, EKS, Lambda, SQS/SNS, CloudFormation, CloudWatch), Google Cloud, Oracle Cloud, Terraform
Platform	Kubernetes, Docker, Helm, Istio, Envoy, gRPC, Jenkins, Buildkite
Observability	Grafana, Prometheus, Datadog, Dynatrace
Verification	Stainless (Scala), Dafny

RESEARCH

Preprints on Zenodo, 2026, each backed by machine-checked proofs in the prime-numbers → project.

Division and Modulo from Recursive Normalization

Using Formal Verification to Prove Properties of Lists Recursively Defined

Formal Verification of Discrete Integration Properties from First Principles

Formal Verification of Cyclic Lists

Formal Verification of Cycle Integral Properties from First Principles

Formal Verification of Euclid's Theorem on the Infinitude of Primes

Formal Verification of Sieve Sequence Stages and Their Transitions

Relaxed Almost-Prime Production in Sieve Sequences

Structural Properties and Signed Boundaries of 2-Gaps in Sieve Sequences

Survival Frontiers in Balanced 2-Gap Companion Processes

EARLIER PUBLICATIONS

Critérios de Usabilidade e Acessibilidade para Avaliação de Aplicações Educacionais 3D na Web — 2015
CBIE-LACLO 2015 · Honourable Mention, Best Published Articles

Definition of Computational Criteria to Evaluation of 3D Anatomical Web Manikins for Teaching in Health Science and for Medical Applications — 2013
PAHCE, Pan American Health Care Exchanges

Accessibility for Web 3D Anatomical Manikins — 2013
SETI, First Symposium on Education and Inclusive Technology

Definition of Criteria for Evaluating 3D Anatomical Manikins on the Web for Education in Healthcare or Medical Applications — 2012
Extensio: Electronic Journal of Extension

SELECTED PROJECTS

Perceptions of Probability 2018–2026

An interactive tool for estimating under uncertainty: it turns optimistic and pessimistic estimates and a plain-language likelihood, such as "probable", into a range instead of a single number.

JavaScript, Statistics Source →

Parallax Engine 2025–2026

A deterministic, testable 3D scene engine for the browser, with real-time head tracking from a webcam that shifts the scene's perspective as the viewer moves.

TypeScript, 3D graphics, computer vision Source →

EDUCATION

Master's, Biomedical Engineering 2011–2013
Universidade de Brasília · four publications · thesis

Graduate Degree, Computer Science 2002–2008
Universidade de Brasília

CERTIFICATIONS AND COURSES

Certificates are available in the courses-and-certifications → repository.

CERTIFICATIONS AND SPECIALISATIONS

AWS Certified Solutions Architect – Associate 2022–2025
Amazon Web Services

Programming with Google Go 2023
Specialisation · University of California, Irvine

Data Science: Foundations Using R 2025
Specialisation · Johns Hopkins University

LEADERSHIP AND MANAGEMENT

Managing Project Risks and Changes 2020
University of California, Irvine

FUNCTIONAL AND PARALLEL PROGRAMMING

Functional Programming in Scala 2016–2024
Courses from the specialisation · École Polytechnique Fédérale de Lausanne

Functional Programming Principles in Scala (2016), Functional Program Design in Scala (2021), Parallel Programming (2024)

Functional Programming with Streams in Java 9 2017
Udemy

CLOUD AND DATA

Introduction to Graph Databases, Cypher, and Neo4j 2020
Pluralsight

AWS Fundamentals: Going Cloud-Native 2019
Amazon Web Services

Leveraging Unstructured Data with Cloud Dataproc 2017
Google Cloud

Big Data and Machine Learning Fundamentals 2017
Google Cloud

Apache Spark 2.0 + Scala: DO Big Data Analytics & ML 2017
Udemy

DATA SCIENCE AND MACHINE LEARNING

Practical Machine Learning 2019
Johns Hopkins University

Regression Models 2019
Johns Hopkins University

Deep Learning in Python 2017
DataCamp

Statistics with R 2016–2023
Courses from the specialisation · Duke University

Introduction to Probability and Data with R (2016), Linear Regression and Modeling (2023)

Introduction to Artificial Intelligence, Advanced Track 2011
Stanford University

OTHER

Smart Contracts with Solidity: Create an Ethereum Contract 2024
Guided project · Coursera

AWARDS

Cannes Lions: Gold, Silver, and Bronze 2014

Gold in Mobile (Media, Entertainment and Leisure), Silver in PR (Use of Digital Platforms), and Bronze in Promo & Activation (New Product Launch) for SAT-JF14, the Record Collection campaign for John Frusciante's album, built at Adjetiva.

Honourable Mention, Best Published Articles 2015
Latin American Conference on Learning Technologies (LACLO)

LANGUAGES

Portuguese (native), English (fluent)